decoherence and branching of the wave function. But there is only one wave function, the wave function of the universe, which describes everything throughout space. How should we think about branching from a global perspective? Does branching happen all at once, or does it gradually spread out from the system where the interaction occurred?”

“Oh boy. I have a feeling this is going to be another unsatisfying answer.” Alice paused to slice off a piece of cheese. She carefully arranged it on a cracker as she thought about the best response. “Basically: that’s up to you. Or, to put the point in more respectable-sounding language, the very phenomenon of ‘branching’ is one that we humans invent to provide a convenient description of a complicated wave function, and whether we think of branching as happening all at once or as spreading out from a point depends on what’s more convenient for the situation.”

Her father shook his head. “I thought branching was the whole point. How can you hold up Many-Worlds as a respectable scientific theory if not only can’t you observe the other branches, and not only can’t you count them, but you don’t even have a definite criterion for how it happens? Branching is just, like, your opinion, man?” He had always been just a little too fond of movie references.

“In a sense, sure. But there are better and worse opinions to have. You may prefer a description in which nothing travels faster than the speed of light. What actually matters is that you can’t communicate or send information faster than light, and that’s true no matter what description you choose to use. But if it makes you feel better to limit an apparently physical effect like branching to propagate no faster than light, you are perfectly welcome to do that. In that case, the number of branches of the wave function would be different depending on where you were in spacetime.” She took out a fresh napkin and began scribbling again, this time making little diagrams out of straight lines. “Here we have space going from left to right and time going upward. Light beams that could potentially be emitted from an event will move upward at forty-five-degree angles. If we start with just a single branch of the wave function, we can imagine branching happening at that event, and then propagating upward in time, but only growing at the speed of light. Observers farther away would be described by a single branch, while nearer ones would be described by two branches. This fits well with the idea that distant observers have no way of knowing, or being influenced by, the branching event, while those nearby do.”